Ex 15: Report and Presentation on Career Pathways and the Future of Low-Code Development

Topic: Patient Appointment Booking System

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1. Aim

To study the career pathways open to low-code developers and the likely future of low-code development, looking at low-code and no-code platforms in general and not only Zoho Creator, and to prepare a report and a presentation outline that link these trends to the skills practised while building the Patient Appointment Booking system in this lab series.

2. Procedure

Step 1: Trace how development has changed — The move from hand coding to AI-assisted building was laid out as a timeline.

Step 2: Compare platforms — Well-known low-code platforms were compared by focus and typical use, to show that the field is wider than one product.

Step 3: Collect current trends — Trends such as AI-assisted building, AI agents, governed citizen development, hyperautomation and legacy modernisation were noted together with their effect on careers.

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Step 4: Map career pathways — Roles were arranged from entry to senior level, with possible switches into other careers.

Step 5: Build a skill stack and roadmap — The skills needed were arranged in layers and turned into a learning roadmap that starts from the experiments already done.

Step 6: Examine risks — Risks such as shadow IT, security, lock-in and the quality of AI-generated output were listed with ways of handling them.

Step 7: Prepare the presentation — A slide-by-slide outline was drawn up so the findings can be presented clearly.

Note: this study rests on general knowledge of the industry. Any statistics or forecasts from articles should be checked against their original sources before being used as facts, and the roles and timeline are a general guide rather than a prediction for any one company.

3. Output

3.1 How Application Development Has Evolved

Each stage reduced the effort needed to produce an application while adding a need for oversight, from hand coding through rapid-development tools and low-code platforms to citizen development and AI-assisted building.

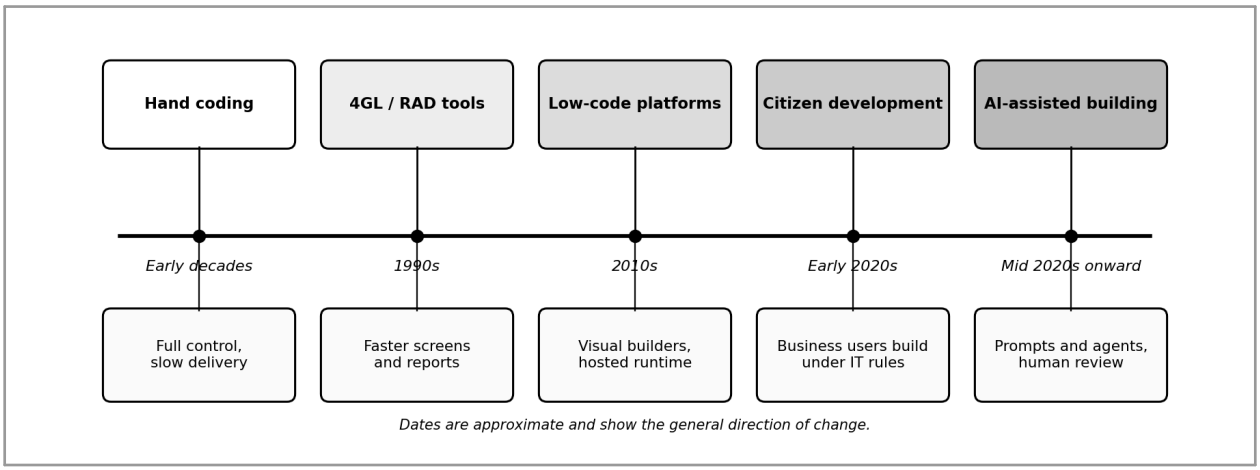


Figure 1: Evolution of Application Development

3.2 The Platform Landscape Beyond Zoho Creator

The platforms below serve different needs. The skills learned on one, such as forms, workflows, data models and APIs, carry over to the others.

Platform	General Focus	Typical Use
Zoho Creator	Business apps with strong links to the Zoho products	Small and mid-size business process apps
Microsoft Power Apps	Apps tied to Microsoft 365 and Azure	Internal tools in organisations that use Microsoft services
OutSystems	Enterprise-scale development with strong code extensibility	Large, complex enterprise applications
Mendix	Enterprise platform with visual modelling	Enterprise and industrial applications
Appian	Process automation and case management	Workflow-heavy, regulated processes
Salesforce Platform	Custom apps built around CRM data	Customer-centred business applications

Table 1: Examples of Low-Code Platforms

3.3 Trends Shaping the Future

Trend	What It Means	Effect on Careers
AI-assisted building	Apps and logic can be drafted from plain-language descriptions	Less routine building, more reviewing, design and testing
AI agents in applications	Agents act inside workflows instead of only suggesting	New roles in designing, limiting and monitoring agents
Governed citizen development	Business users build apps within IT rules	Demand for platform owners who set standards
Hyperautomation	Low-code, RPA and AI combined to automate whole processes	Automation engineers and process designers are valued

Trend	What It Means	Effect on Careers
Legacy modernisation	Old systems are wrapped or replaced with low-code	Integration and migration skills gain importance
Fusion teams	Business and technical staff build together	Communication and requirement skills matter as much as tools

Table 2: Trends and Their Effect on Careers

3.4 Career Pathways

Common routes lead from entry roles such as citizen developer or junior low-code developer, through development, integration and automation roles, to senior roles such as architect or platform lead. Dashed arrows show switches into pro-code engineering or product and business roles.

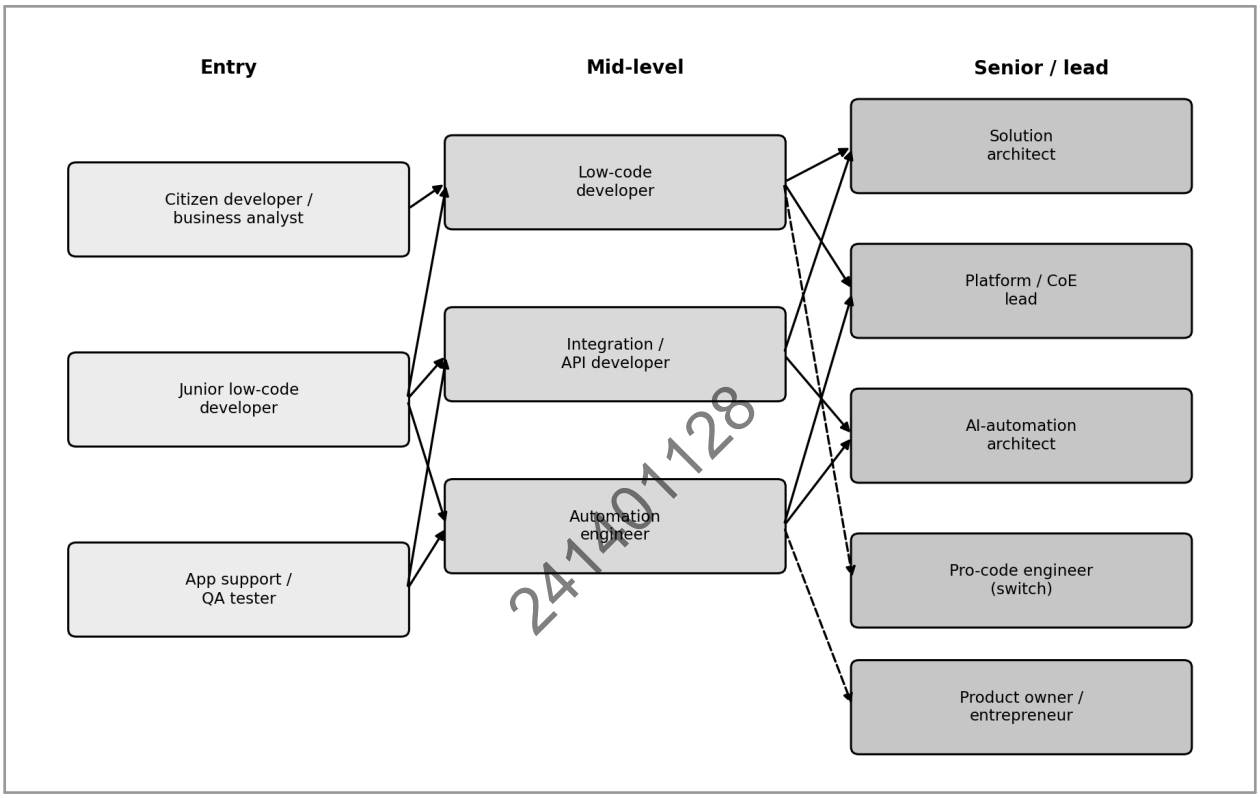


Figure 2: Career Pathways in Low-Code Development

Role	What the Person Does	Key Skills
Citizen developer / analyst	Builds simple apps for their own team and gathers requirements	Process thinking, basic platform use
Low-code developer	Builds forms, workflows, reports and scripts	Platform skills, data modelling, scripting
Integration / API developer	Connects apps to other systems through APIs	REST, JSON, authentication, error handling
Automation engineer	Automates end-to-end processes, including AI steps	Workflow design, RPA tools, AI services
Solution architect	Designs the overall solution, security and governance	Architecture, security, platform strategy
Platform / CoE lead	Sets standards and supports citizen developers	Governance, training, communication

Table 3: Roles at a Glance

3.5 Skill Stack for a Low-Code Career

Skills build in layers from foundations such as logic and data up to governance. The upper layers are the least likely to be replaced by automation.

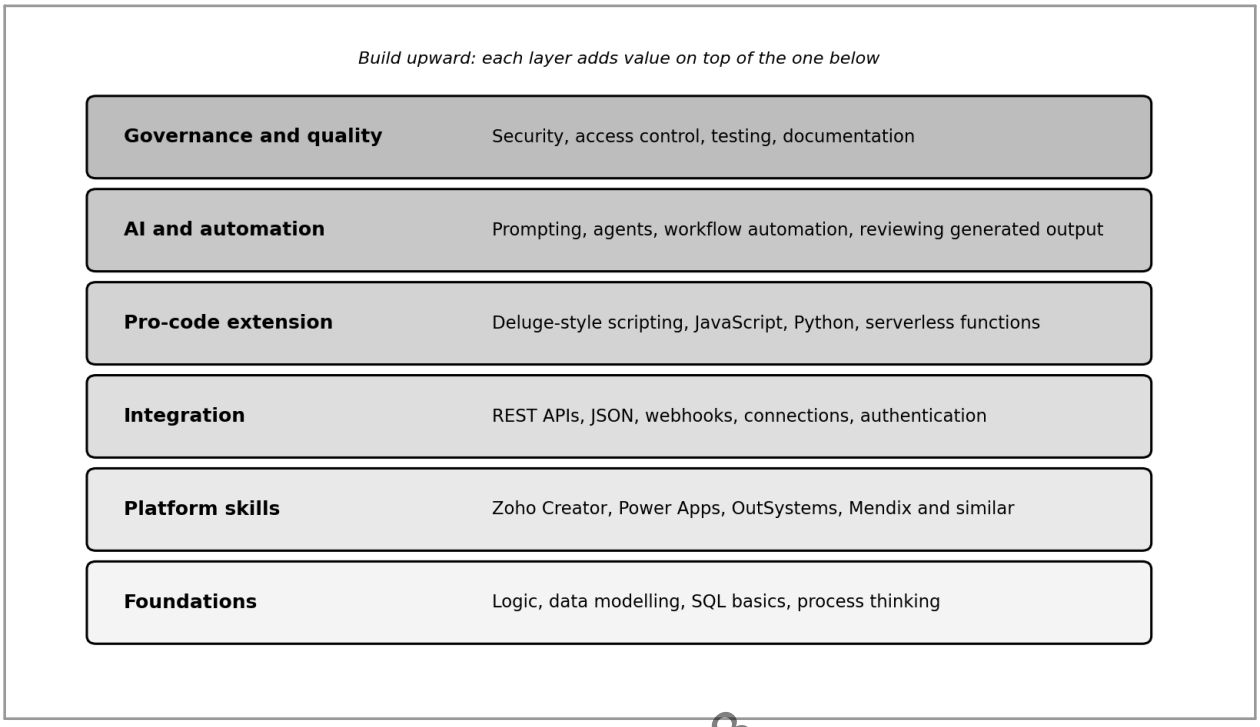


Figure 3: Skill Stack for a Low-Code Career

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3.6 Learning Roadmap

The roadmap moves from basics to projects, then to connections and scripting, AI skills and finally specialisation. The experiments in this lab series sit mainly in steps two and three, from the booking form, reports and dashboard to Deluge workflows, the CRM integration attempt and the pro-code extension.

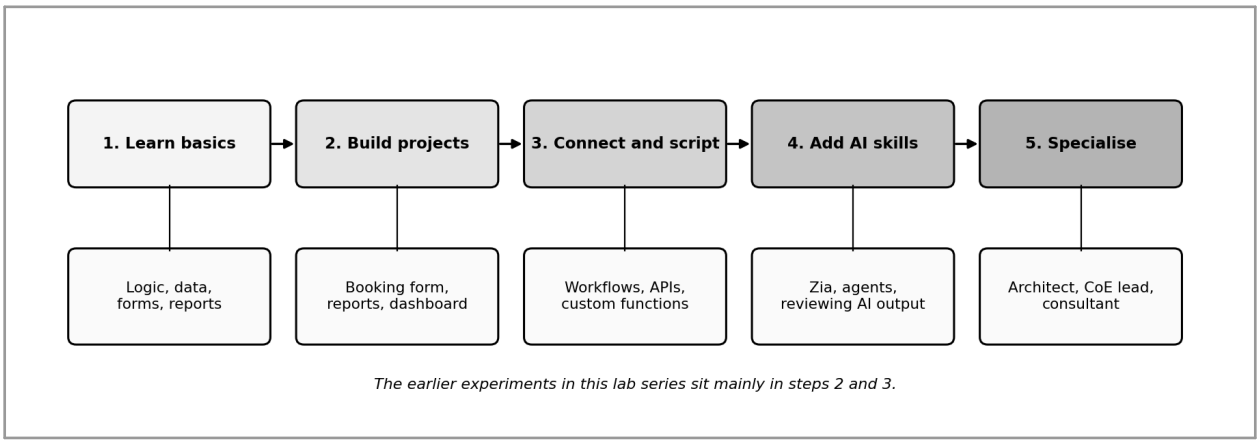


Figure 4: Learning Roadmap

3.7 Risks and Challenges

Challenge	Why It Matters	How to Handle It
Shadow IT	Apps built outside IT may be unknown, unsecured or duplicated	Central inventory, approval process, centre of excellence
Security and compliance	Easy building can hide weak access control and data exposure	Role-based access, reviews, testing and audit trails

Challenge	Why It Matters	How to Handle It
Quality of AI-generated output	Generated logic can contain errors that look correct	Human review, test cases and version control
Vendor lock-in	Apps depend on one platform's tools and pricing	Keep logic in APIs, keep data exportable, learn transferable skills
Changing job market	Routine build tasks may shrink as AI improves	Move toward design, integration, governance and AI oversight

Table 4: Risks and How to Handle Them

3.8 Likely Future Workflow

In the likely future workflow a business need is described in plain language and AI drafts the application. People review the logic, data and security, the app is released through governance, and monitoring feeds new improvements. Human judgement stays at the review and governance steps.

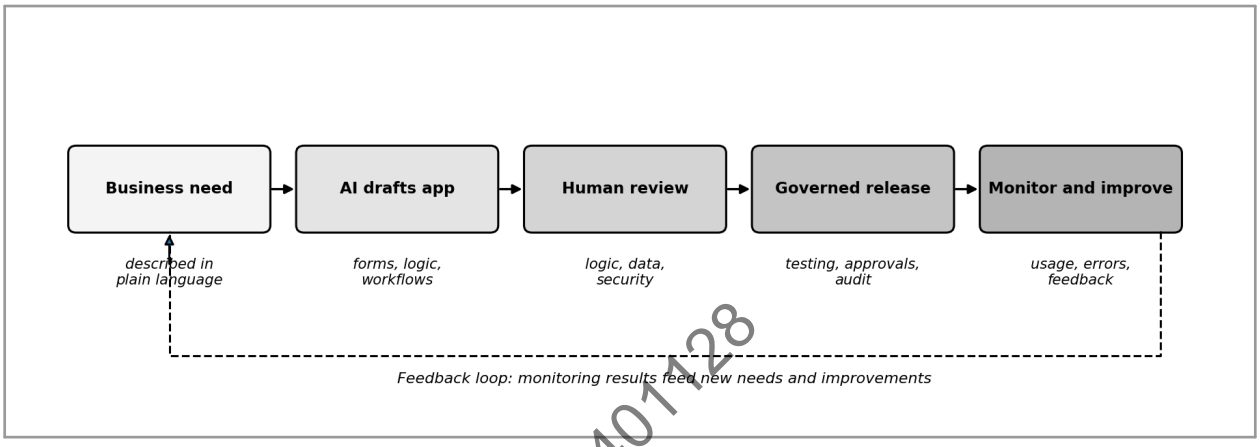


Figure 5: Human and AI Building with Governance

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3.9 Presentation Outline

The findings can be presented in ten slides, with one idea on each:

Slide	Title	Key Points
1	Title and introduction	Topic, name and register number
2	What is low-code?	Visual building with the option to add code; the appointment booking system as an example
3	How development evolved	Timeline from hand coding to AI-assisted building
4	Platform landscape	Zoho Creator, Power Apps, OutSystems, Mendix, Appian, Salesforce
5	Future trends	AI-assisted building, agents, governed citizen development, hyperautomation
6	Career pathways	Pathway diagram from entry to senior roles
7	Skills and roadmap	Skill stack and learning roadmap
8	Risks and challenges	Shadow IT, security, AI output quality, lock-in
9	My experience	The booking app, workflows, reports, integration attempt, security plan and pro-code extension
10	Conclusion and questions	Low-code and pro-code will work together; keep learning

Table 5: Slide-by-Slide Outline

4. Result

The study showed that low-code is moving from a quick way of building simple apps to a wider field in which visual building, pro-code extension and AI work together. The platform landscape is broad, and skills learned on one platform such as Zoho Creator transfer to others.

Career pathways run from citizen developer and junior low-code developer, through integration and automation roles, to architect, platform lead and AI-automation roles, with routes into pro-code engineering and product work. The skill stack and roadmap show where the Patient Appointment Booking experiments fit on that path. The main risks are shadow IT, security, the quality of AI-generated output and lock-in, which are managed through governance and human review. The report and the slide outline are ready to be turned into a presentation.

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